

Worksheet 7 *Phase portraits*

Prepares you for lab manual pages **Ph1, Ph2, Ph3, Ph4** (section L3 S2)

Name:

Name of TA:

1. Suppose that A is a quantity that obeys the change equation

$$A' = 2A(3 - A)(A - 5).$$

- (a) Draw a phase portrait for A below. Shade in the circles or leave them empty, to indicate stable or unstable equilibria.

_____ → A

- (b) Sketch three time series for A , with starting values $A = 1$, $A = 4$, and $A = 6$. Highlight the equilibrium points with dashed horizontal lines.



2. Suppose that B is a quantity that obeys the change equation

$$B' = \frac{2B(4 - B)}{1 + B^2}.$$

- (a) Draw a phase portrait for B below. Shade in the circles or leave them empty, to indicate stable or unstable equilibria.

_____ → B

- (b) Describe in words what will happen in the long term if B starts at $B = 3$ at time zero.